

EXERCISE 1
Hypothesis Testing for 2 Sample



SECI2143
PROBABILITY & STATISTICAL DATA ANALYSIS

(i) James and his sister, Alison, each deliver 30 newspapers and each are paid the same amount. Alison claims that the equal payment is unfair as she takes on average longer to distribute the newspaper compare to her brother, James. To test her claim, her father records their delivery times (in minutes) as shown below.

Table 1: Delivery times (minutes)

James	Alison
45	49
30	42
39	39
32	45
34	38
43	49
38	43
35	36
43	33
39	41
32	36
34	-
$\bar{x}_1 = 30$ $s_1 = 4.92$	$\bar{x}_1 = 33$ $s_1 = 5.25$

a. What is the assumption required for this hypothesis analysis? (1 mark)

b. State the null and alternative hypotheses. (2 marks)

c. Variance of both population are unknown and assume both are similar, test whether Alison's claim is justified, using a 5% level of significance. (7 marks)

d. Based on result in (c), what change would you make to your test procedure so that the conclusion of the hypothesis is different than (c)? (2 marks)

a) As a part of an industrial training program, some trainees are instructed by Method A, which is straight teaching-machine instruction, and some are instructed by Method B, which also involves the personal attention of an instructor. Random samples of size 10 are taken from large groups of trainees instructed by each of these two methods, and the scores they obtained in an appropriate achievement test are shown in Table 1.

Table 1: Scores of trainees

Method A	71	75	65	69	73	66	68	71	74	68
Method B	72	77	84	78	69	70	77	73	65	75

Given: $s_A^2 = 11.3333, \quad s_B^2 = 29.1111$

Use 0.05 level of significance to test the claim that Method B is more effective. Assume that the populations sampled can be approximated closely with Normal distribution having the same variance.

THANK YOU